

# ***Internal Fault Detection in Distribution Transformers***



**FMEA Energy Connections Conference**

**November 6, 2015**

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# Why is internal fault detection important?

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# Why internal fault detection is important



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## ***Linemen Safety***

- ☐ Operating challenges
- ☐ Crew pressures
- ☐ Risk every time we re-close

## ***Troubleshooting Effectiveness***

- ☐ Speed
- ☐ Accuracy

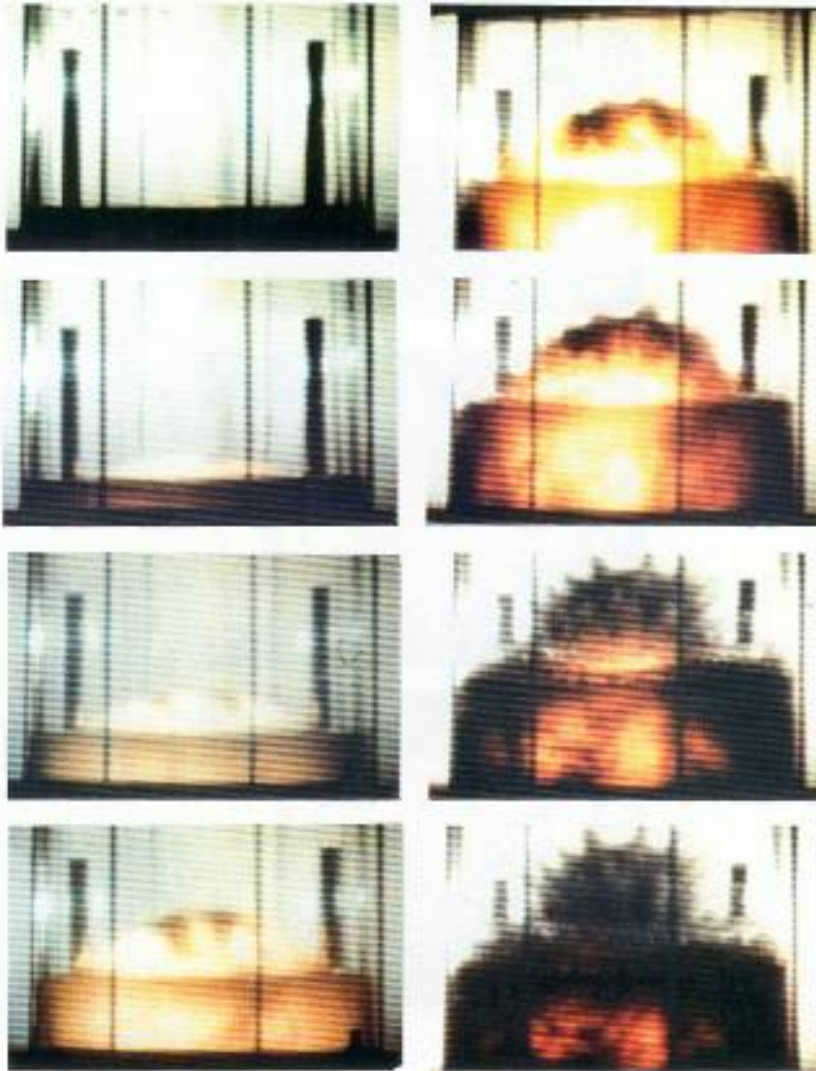
## ***System Reliability***

- ☐ Reduce outage times
- ☐ SAIDI/SAIFI

# Why internal fault detection is important



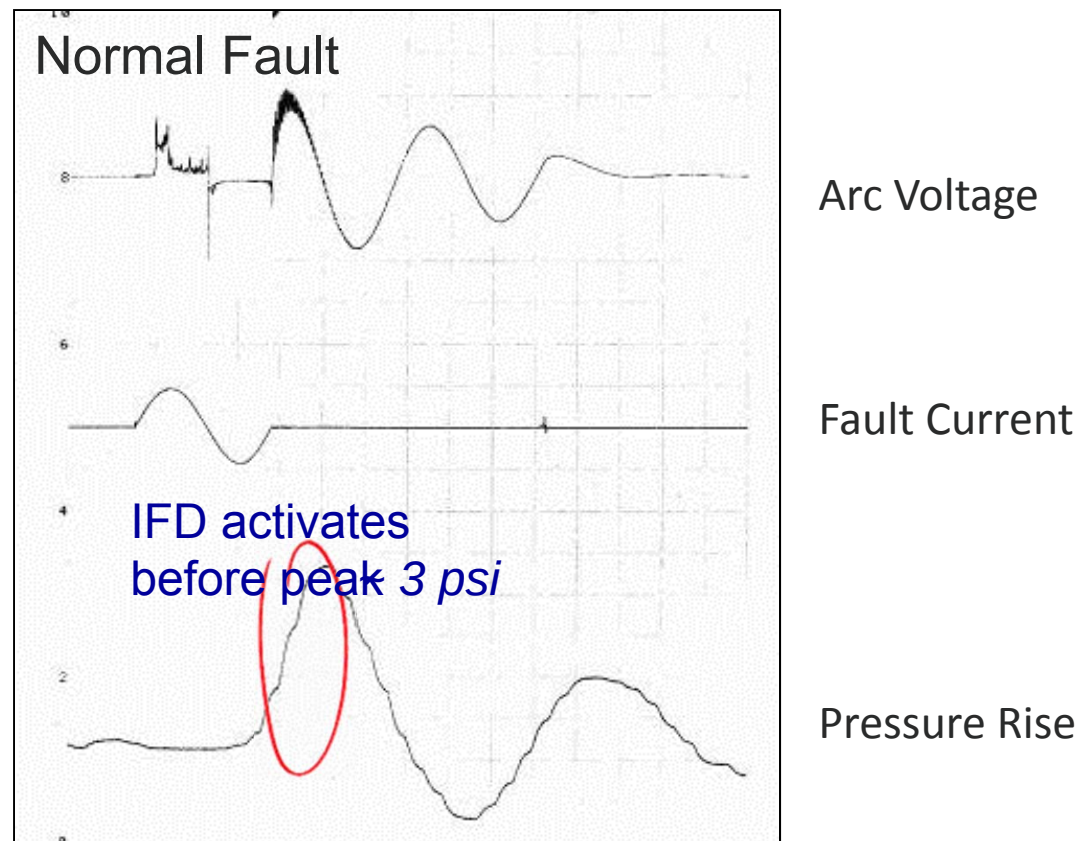
# What does an internal transformer fault look like?



- A rapid, transient pressure rise occurs in every internal arcing fault
- Variable peak pressure
- Consistent rate of rise

# What does an internal transformer fault look like?

*500A test*



*Certification tests at Powertech Labs*

# What about PRV's?

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- PRV operation:  
 $P > 10 \pm 2$  psi  
 $\neq$  Internal fault

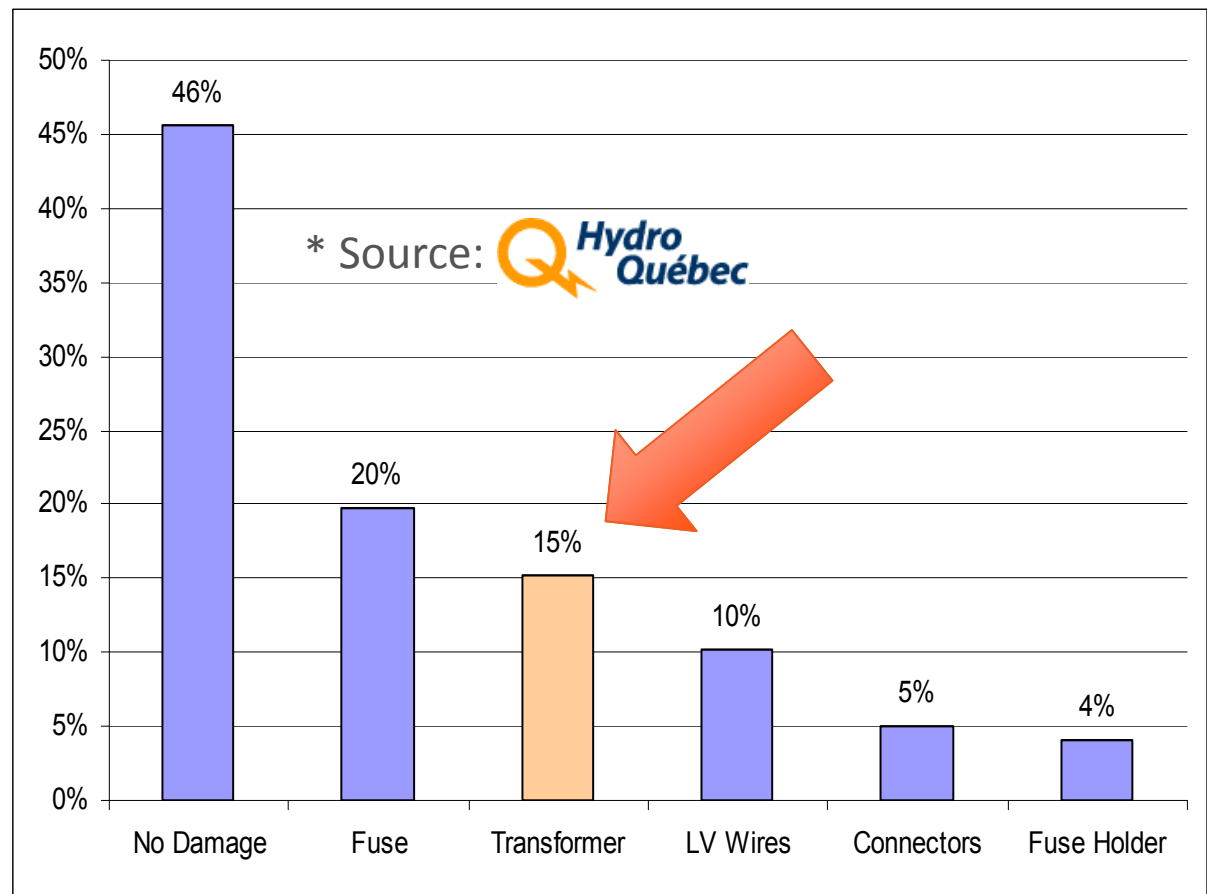
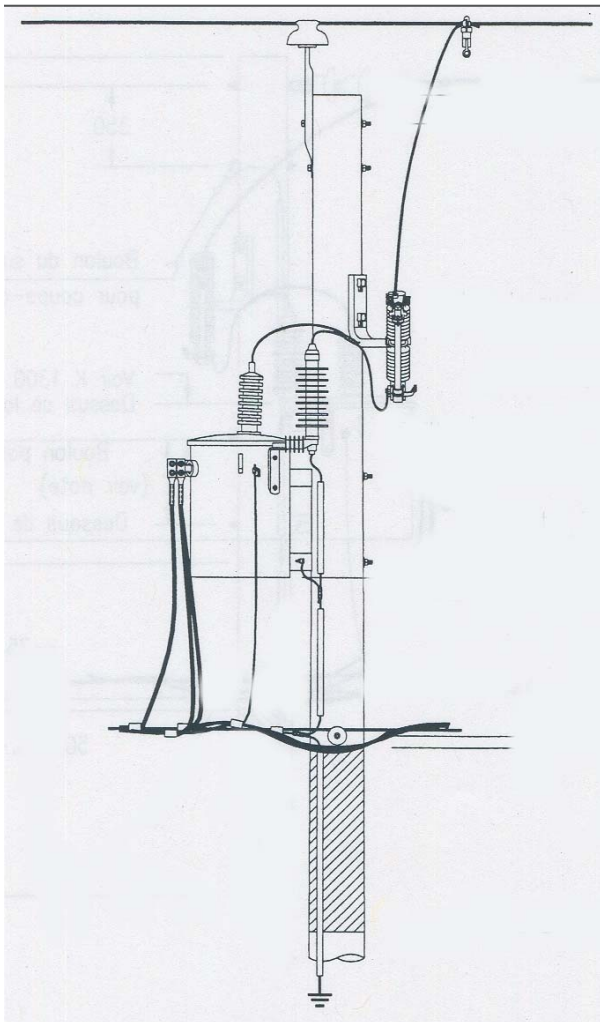
- (Poppet style PRV) 'does not have time to react to the overpressure rates of rise and values produced when low-impedance faults occur.'
- A. Even et al., "Safety of Distribution Transformers against internal failure", Conference Publication No. 438., IEEE, 1997



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**What causes transformer fuses to blow?**

# Causes of transformer fuse operations



# Causes of padmounted transformer fuse operations

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- Internal Transformer Faults
- Secondary Faults
- Overloads
- Oil Temperature
  - (Dual Element Fuse)



# Current Industry Practices

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1) Automatic Replacement

2) Field Testing

3) Trial & Error



# Sample utility procedures

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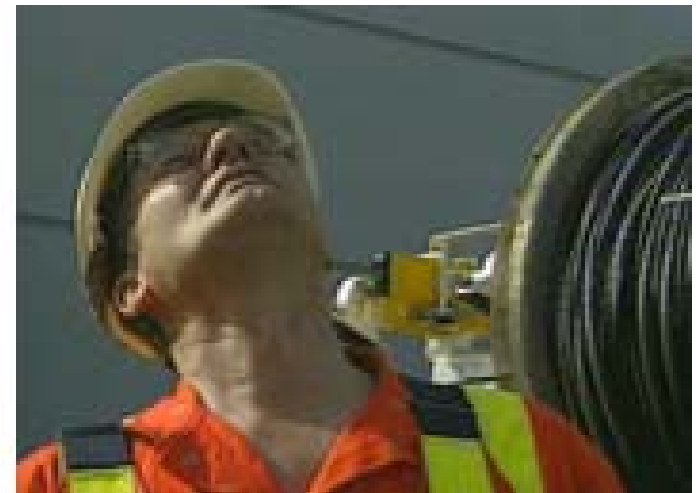
- 1) Always wear the appropriate personal protective equipment including fire retardant clothing, hard hats, safety glasses and rubber gloves.
- 2) When performing any close inspections of or repairs to the transformer always deenergize the unit and take precautions from any sources of power including customer generators.
- 3) Prior to re-fusing transformers, make a thorough inspection of the transformer and the surrounding area, looking for indicators such as: animal carcasses, bulged tank or cover, discolored tank, oil leak, burned oil aroma, flashed or broken bushings, any short circuits such as wrapped wires in the secondary or service.
- 4) Prior to re-fusing transformers, test the transformer with an approved instrument such as a Transformer Turns Ratio tester. Some companies require some form of testing on every transformer before they are energized.
- 5) When re-fusing a suspected faulty transformer, position yourself as far away as possible by using an extendable live line tool. Some companies specify maintaining a distance of at least 10 ft. when re-fusing transformers.
- 6) When re-fusing a suspected faulty transformer, always disconnect the customer's load.
- 7) Test the transformer with a smaller test fuse. (Some companies have test fuse tables based on the transformer size and voltage).
- 8) If the transformer blows a fuse and the pressure relieve valve has operated, do not re-fuse and proceed with replacement.
- 9) If the transformer has an auxiliary current limiting fuse that has blown, do not re-fuse; proceed with replacement. Checking transformer condition
- 10) Maintain proper distance when re-fusing. In this condition: do not refuse
- 11) Always make sure the neutral is connected first before connecting a transformer and that it is removed last if disconnecting a transformer.
- 12) Do not allow the paralleling of transformers across any point that might be used to isolate a line section, such as switches, disconnects and double dead ends to avoid the possibility of back feed into a cleared line section.
- 13) Always wear rubber gloves when working on the secondary side of a suspected faulty transformer.
- 14) Take precautions when removing the lid from a suspected faulty transformer by operating the pressure relief valve. If it is an older unit without a valve, tie a sling over the lid before loosening the attachments.
- 15) Some companies have established step-by-step procedures for investigating transformer problems while others allow the lineworker to access the situation and take precautions they deem necessary within mandatory safety guidelines.
- 16) It is recommended that step-by-step procedures be developed as they are good training aids and can serve as a refresher for lineworkers. They also help ensure certain key steps are followed

# Sample utility procedures

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OR



Look Up

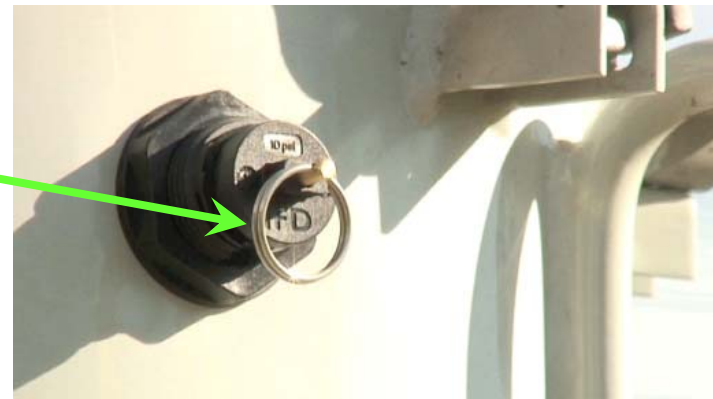
# Overview :: 2 functions

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1. **Internal arcing fault detection** Flag comes out when internal fault has occurred

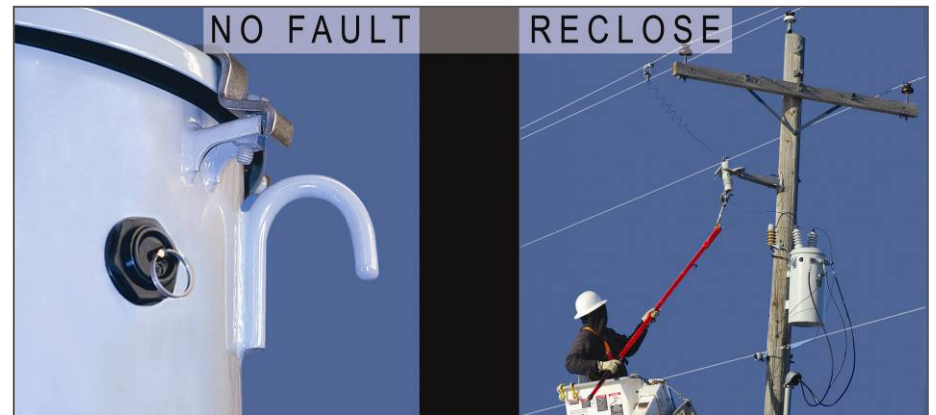
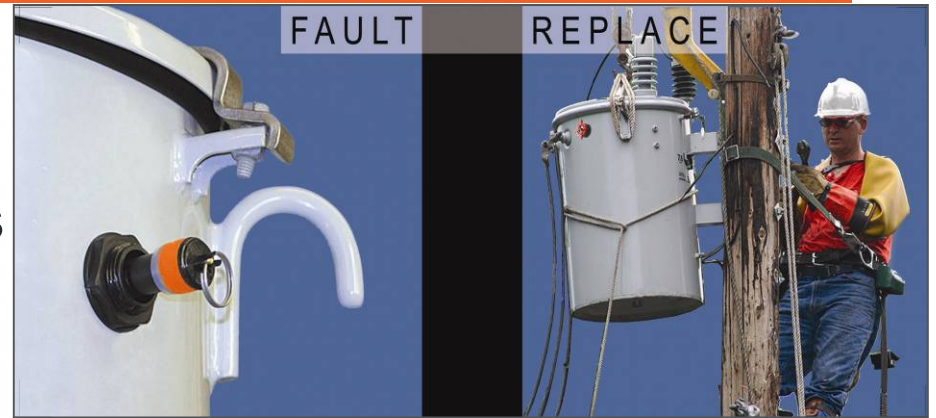


2. **Pressure relief device**  
Pull the ring to operate manually

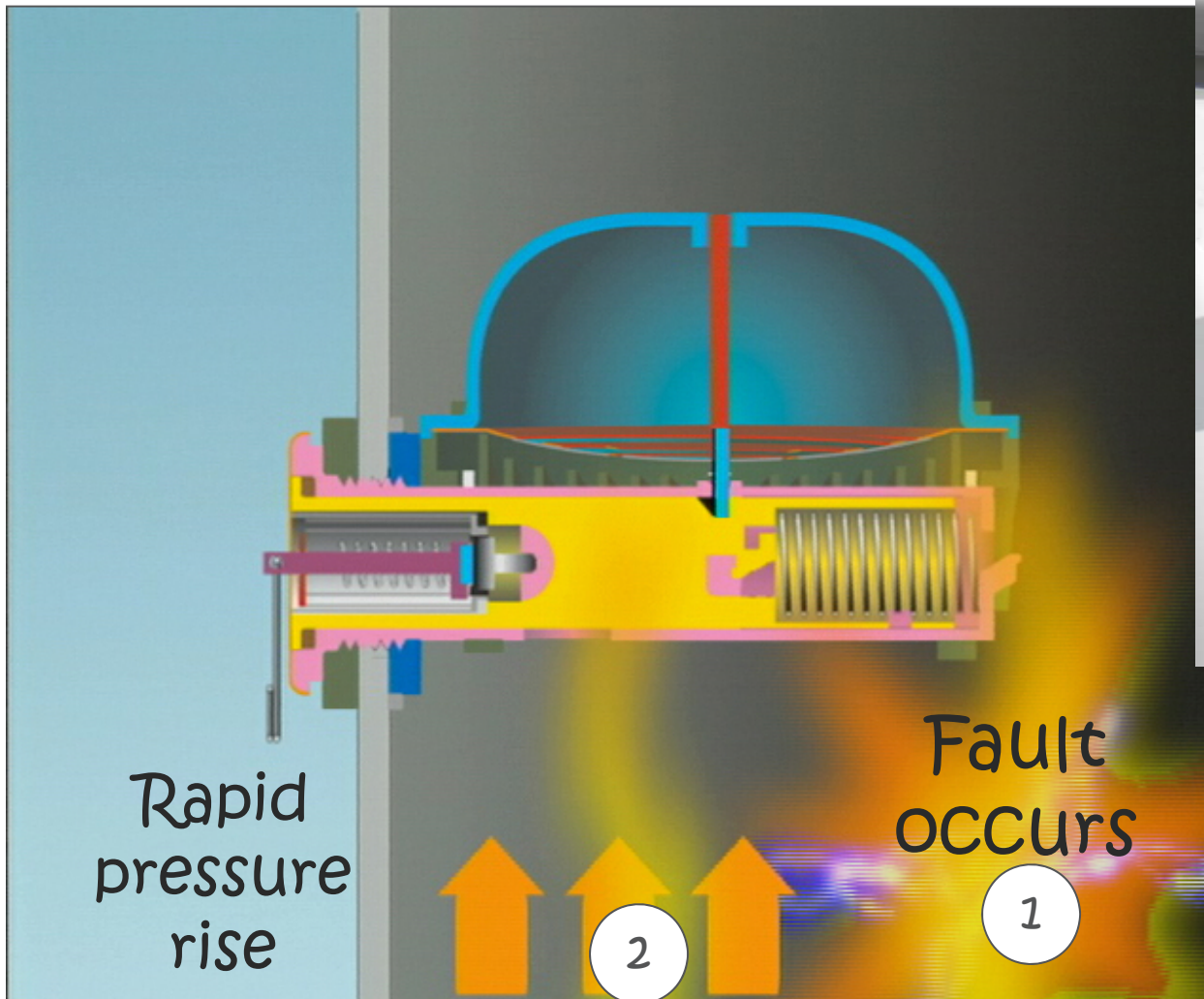


# New line crew practices

- ❑ If the IFD has activated, replace the transformer
  - it has failed internally and is dangerous to re-energize
  
- ❑ If fuse cutout is open and the IFD has not activated, likely OK
  - still conduct all your normal safety check procedures



# How it Works



IFD activates

# One-Size-Fits All

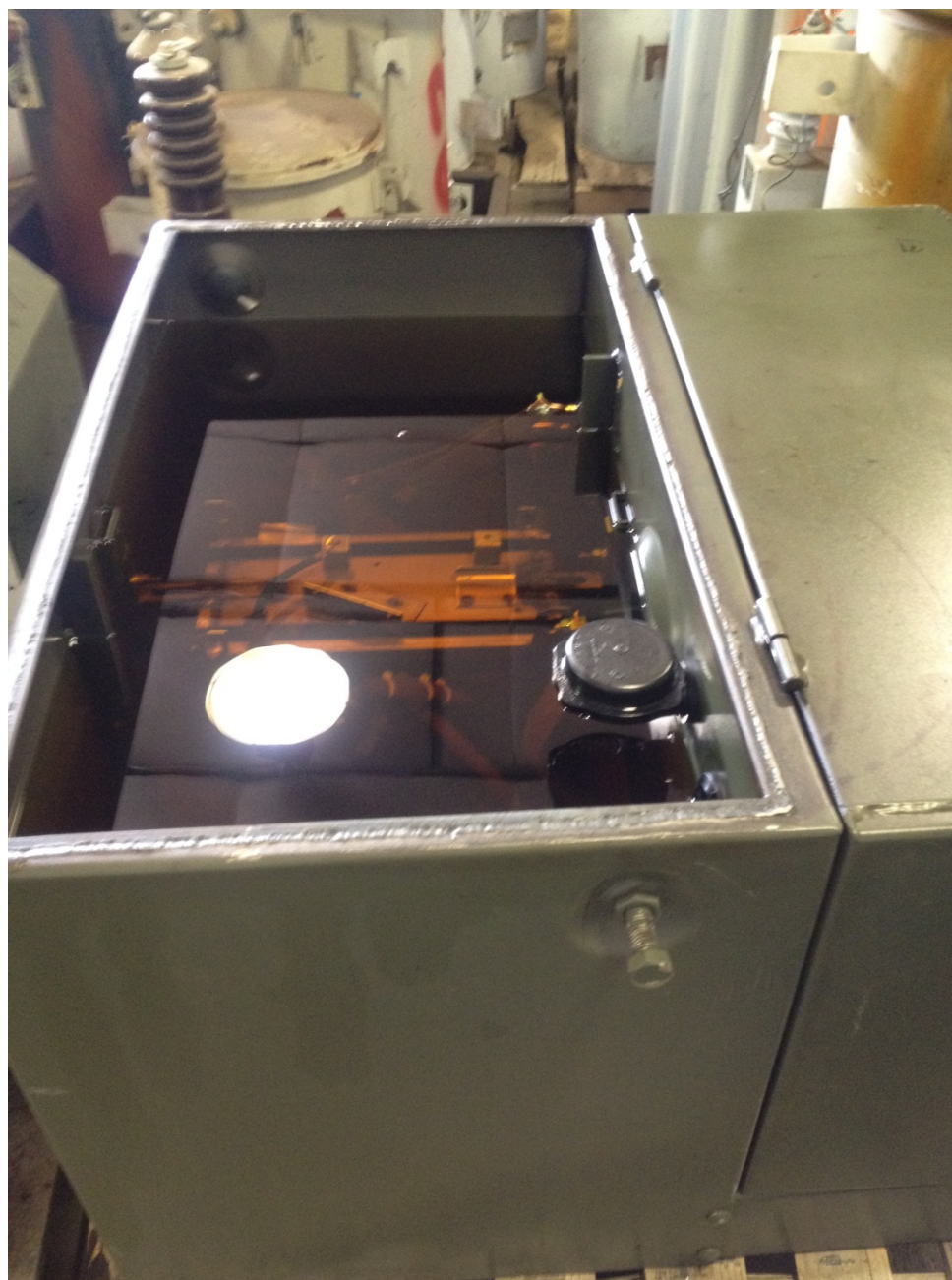


# Installation

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- ❑ Air space, above oil
- ❑ Process adapted to each transformer manufacturer







© IFD Corporation  
2001-2011

# 3-Phase Overhead Banks

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- ❑ Can be difficult to troubleshoot
  - Variety of connection types
  - Access/clearance issues
  
- ❑ The IFD speeds up the process
  - Looks only at sudden pressure
  - Independent of connection types or any other electrical considerations

# IFD status today

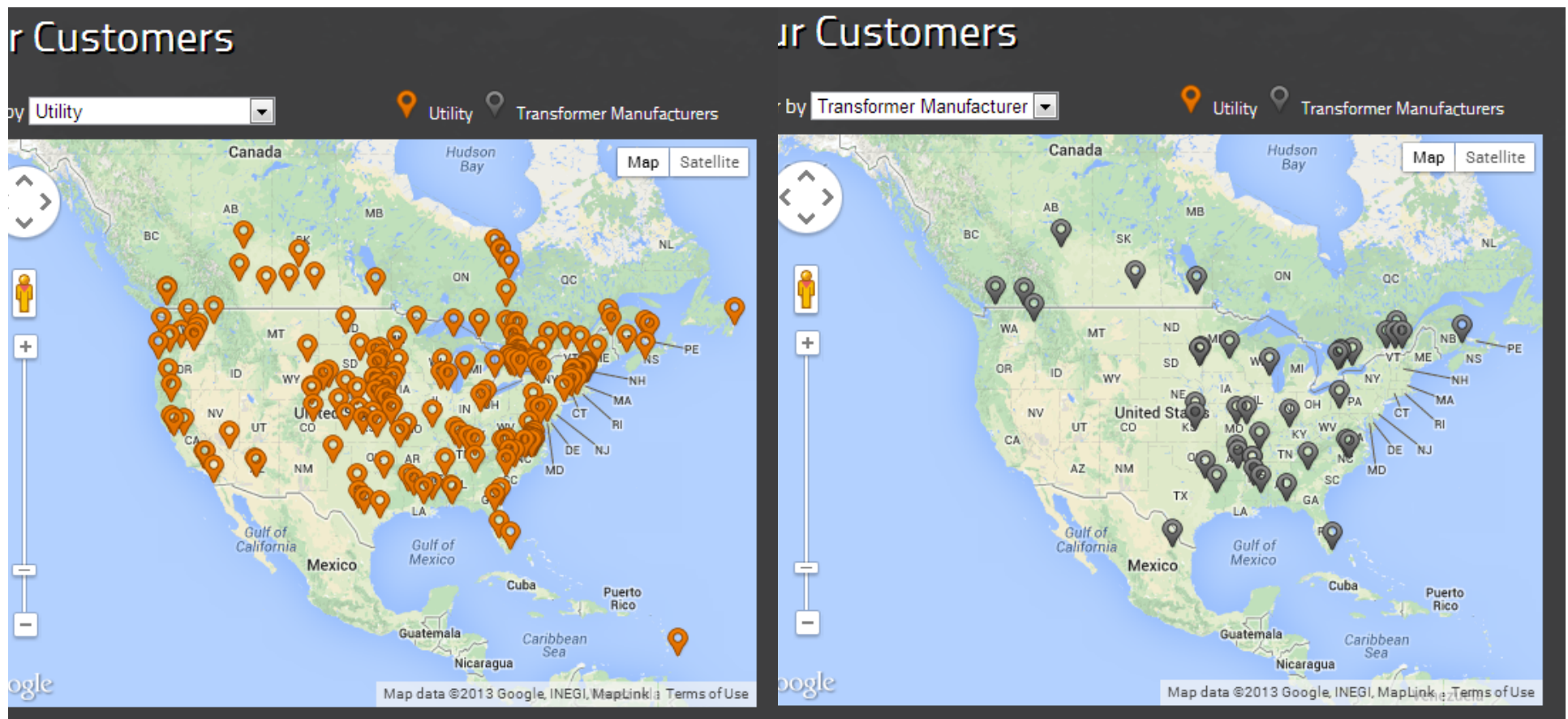


- ☐ Installed since 2001
- ☐ Over 850,000 in service
- ☐ ~ 400 Utility Users
- ☐ ~ 50 Installers
- ☐ Canada: 85% usage rate

- |                                                     |                                                      |
|-----------------------------------------------------|------------------------------------------------------|
| <input checked="" type="checkbox"/> ABB / PPI       | <input checked="" type="checkbox"/> Moloney Electric |
| <input checked="" type="checkbox"/> Central Moloney | <input checked="" type="checkbox"/> SE Transformer   |
| <input checked="" type="checkbox"/> Howard          | <input checked="" type="checkbox"/> Siemens          |
| <input checked="" type="checkbox"/> ERMCO           | <input checked="" type="checkbox"/> Carte            |
| <input checked="" type="checkbox"/> Cooper          | <input checked="" type="checkbox"/> CamTran          |
| <input checked="" type="checkbox"/> GE Prolec       | <input checked="" type="checkbox"/> PTI              |
| <input checked="" type="checkbox"/> Solomon         | <input checked="" type="checkbox"/> CG Power Systems |
| <input checked="" type="checkbox"/> Emerald         | <input checked="" type="checkbox"/> Others           |

# IFD customers

□ <http://ifdcorporation.com/product/our-customers/>



*“...this is great, a safety device that makes us money.”*

Vice-President T&D

## Value Summary

|                                                                          | Pole mounted                                               | Pad mounted                                                |
|--------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|
| Typical cost per transformer                                             | <b>\$1,000</b>                                             | <b>\$10,000</b>                                            |
| <b>Efficiency</b> (Trouble call time savings)<br>1-phase<br>3-phase bank | <b>\$120 – 350</b><br><b>\$480 – 960</b>                   | <b>\$400 – 800</b><br><b>\$400 – 800</b>                   |
| <b>Materials Saved</b><br>Fuses                                          | <b>\$20 – 30</b>                                           | <b>\$20 – 30</b>                                           |
| <b>Diagnostics</b><br>Keep good transformers in service                  | <b>\$3,000 – 4,000</b>                                     | <b>\$5,000 – \$15,000</b>                                  |
| <b>Reliability</b><br>Improve SAIDI / SAIFI                              | <b>~ \$200,000 / yr</b>                                    | <b>~ \$200,000 / yr</b>                                    |
| <b>Cost Avoidance</b><br>Environmental<br>Worker / public safety         | <b>\$1,000 – 60,000</b><br><b>\$ thousands to millions</b> | <b>\$1,000 – 60,000</b><br><b>\$ thousands to millions</b> |

# Sample Specification Wording

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“Each transformer shall be equipped with a non-resettable device which detects and provides external indication of internal transformer faults, and also incorporates pressure relief functionality. The approved device is manufactured by IFD Corporation or approved equal.”



## THE GOAL

- ☐ Improve linemen safety
- ☐ Enable faster and more accurate decision making
- ☐ Provide condition status information for the entire life of the transformer
- ☐ Keep unfaulted units in service
- ☐ Improve system reliability